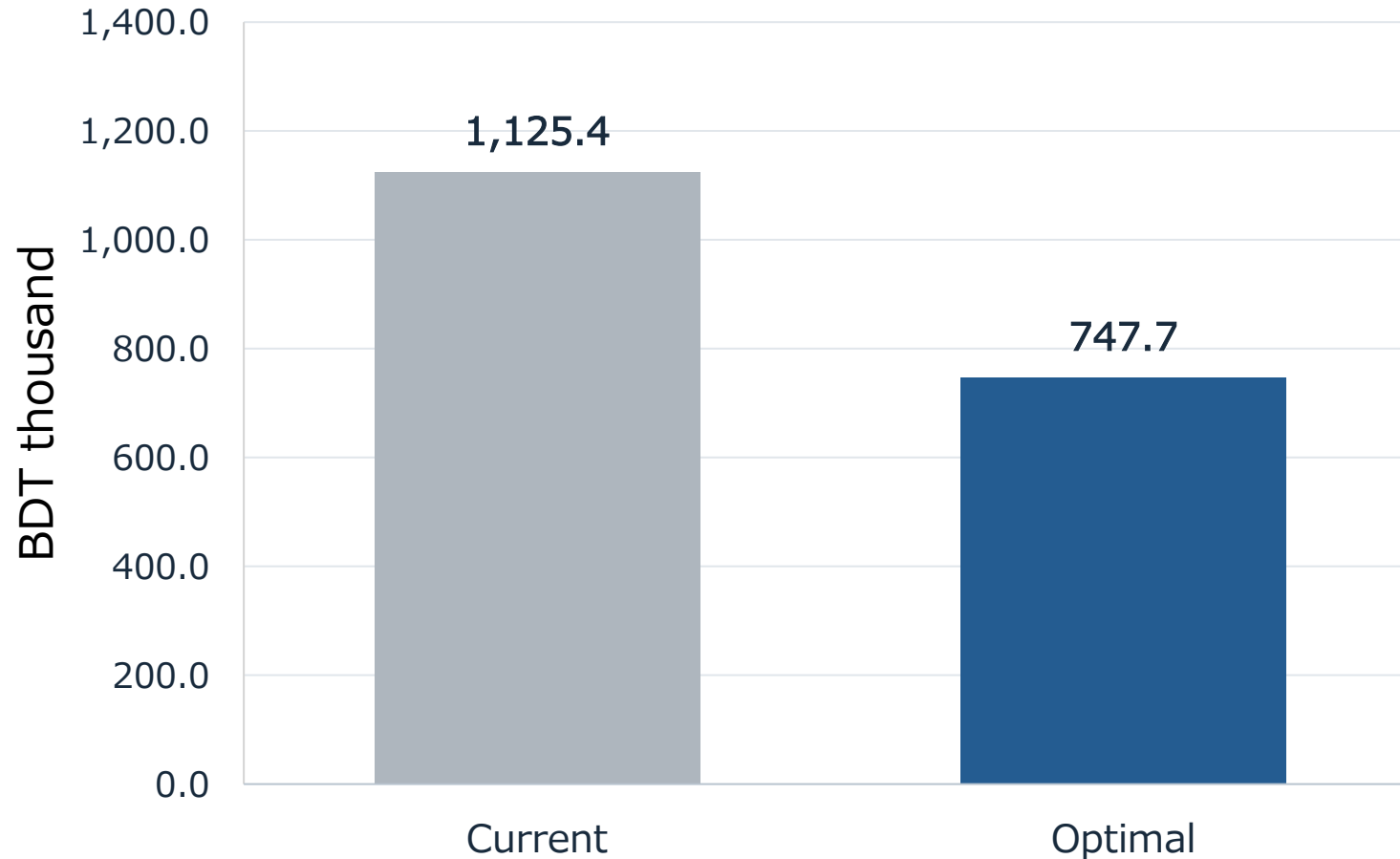


Chips Production Cost Reduction Production Planning Optimization

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Bangladesh FMCG chips plant case study

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BDT 377,684.58

Modeled monthly savings

253 MT of demand met

2 active lines

3 SKUs supplied

Savings depend on the supplied costs, demand and operating assumptions.

The model meets monthly demand and reduces production-related cost by 33.56%.

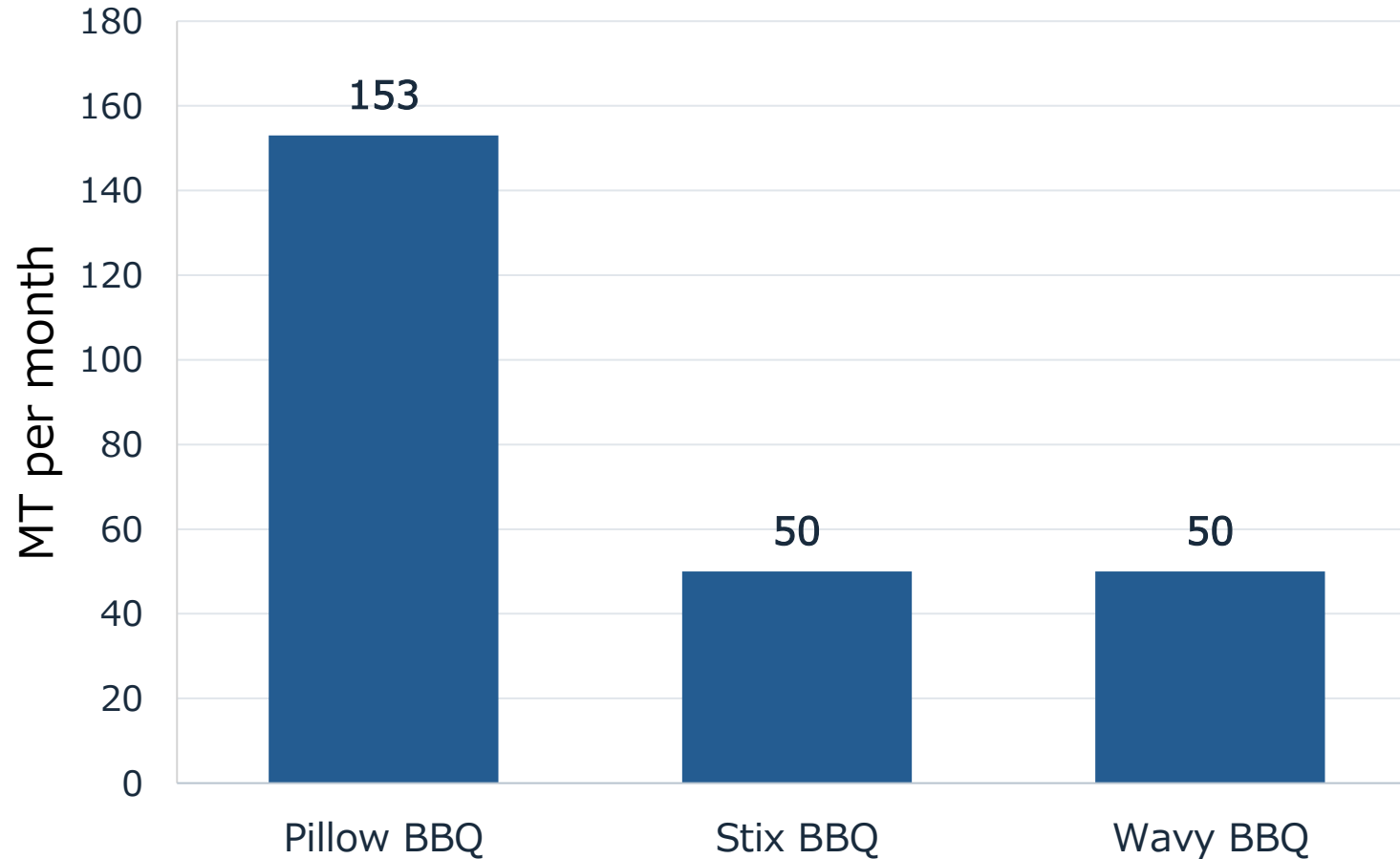
Part	Main question	Report content
Plant context	What must the plant produce?	Demand, costs, capacities and operating rules
Optimization model	How does the plan minimize cost?	Decision variables, objective and constraints
Results and sensitivity	What changes and how robust is it?	Allocation, savings and scenario analysis
Execution and appendix	What is needed for a plant pilot?	Model improvements, KPIs and calculations

The report connects plant inputs, mathematical decisions and a proposed implementation plan.

Planning input	Case assumption
Products	Pillow BBQ, Stix BBQ and Wavy BBQ
Available equipment	Line 1, Line 2 and Line 3
Calendar	25 working days per month, 2 shifts per day, 8 hours per shift
Operating restrictions	At most 2 active lines and at most 2 SKUs per line per day
Cost scope	Production-related variable costs and line activation costs

The objective treats line fixed costs as monthly activation costs. Confirm their avoidability before a pilot.

The case plans a representative production day across a 25-day month.



253 MT/month

Daily requirement

Pillow: 6.12 MT

Stix: 2.00 MT

Wavy: 2.00 MT

Total: 10.12 MT

Pillow BBQ accounts for 60.47% of monthly volume and drives the line allocation.

Variable production-related cost, BDT per MT

Line	Pillow BBQ	Stix BBQ	Wavy BBQ
Line 1	2,340.91	1,754.68	2,783.78
Line 2	1,523.64	1,457.39	2,394.29
Line 3	2,340.91	1,754.68	2,783.78

Monthly fixed cost, BDT: Line 1 = 102,375 / Line 2 = 204,750 / Line 3 = 102,375

Line 2 offers the lowest variable cost for every SKU but carries twice the fixed cost.

SKU-specific capacity, MT per day at 2 shifts

Line	Pillow BBQ	Stix BBQ	Wavy BBQ
Line 1	4.40	5.87	3.70
Line 2	11.00	11.50	7.00
Line 3	4.40	5.87	3.70

At most 2 of 3 lines operate per day.
Each active line can produce at most 2 SKUs per day.

SKU capacities are alternative full-day rates. A shared-time check appears later in the report.

The model combines SKU capacity limits with a two-line and two-SKU operating rule.

Cost element	Calculation / definition	BDT per month
Product wastage	Reported amount	121,875
Operator salary	$8 \times \text{BDT } 29,000$	232,000
Servicing	Reported amount	15,000
Fixed subtotal	$121,875 + 232,000 + 15,000$	368,875
Variable cost	Combined labor and electricity	756,500
Total baseline	$368,875 + 756,500$	1,125,375

The source does not provide a baseline dispatch plan or evidence that every fixed cost can be avoided.

The reported baseline totals BDT 1,125,375 per month.

Variable	Meaning	Type	Count
$Q(s,l)$	Daily quantity of SKU s on line l , MT	Continuous, ≥ 0	9
$y(l)$	Whether line l operates	Binary, 0 or 1	3
$z(s,l)$	Whether SKU s may run on line l	Binary, 0 or 1	9
Total	3 SKUs $\times$ 3 lines plus line activation	MILP model	21

Mixed-integer linear programming (MILP)

Binary decisions represent the choice of lines and SKU assignments.

The formulation has 9 continuous quantities and 12 binary decisions.

$$\text{Minimize } Z = 25 \times \sum_s \sum_l [C(s,l) \times Q(s,l)] + \sum_l [F(l) \times y(l)]$$

Term	Unit	Economic meaning
$25 \times C(s,l) \times Q(s,l)$	BDT per month	Quantity follows the cheapest feasible line allocation
$F(l) \times y(l)$	BDT per month	A line adds fixed cost when the model activates it
Z	BDT per month	Total production-related cost within the stated scope

The fixed-cost interpretation follows the implemented monthly objective.

The objective adds monthly variable production cost and active-line fixed cost.

Constraint	Formulation	Purpose
Demand, 3 rows	$25 \times \sum_l Q(s,l) \geq D(s)$	Meet each SKU's monthly requirement
SKU capacity, 9 rows	$Q(s,l) \leq P(s,l) \times z(s,l)$	Limit production for each SKU-line pair
Line activation, 3 rows	$\sum_s Q(s,l) \leq 25 \times y(l)$	Force a line to activate if it produces
Operating limit, 1 row	$\sum_l y(l) \leq 2$	Use at most 2 lines
SKU limit, 3 rows	$\sum_s z(s,l) \leq 2$	Assign at most 2 SKUs to each line
Variable domains	$Q \geq 0$ and y, z binary	Define feasible decision types

The value 25 in the activation constraint is an artificial big-M bound, separate from the 25-day calendar.

The notebook enforces demand, SKU capacity and line-use limits through 19 linear constraints.

Step	Notebook activity	Output / control
1. Load inputs	Define cost, capacity and demand values	Consistent BDT, MT and monthly/daily units
2. Formulate	Create Q, y and z variables, objective and constraints	21 variables and 19 explicit constraints
3. Solve	Call PuLP CBC solver	Recorded status: Optimal
4. Interpret	Read quantity and line-use values	Production allocation and BDT 747,690.42 cost
5. Test scenarios	Change demand, capacity or variable cost	Sensitivity results and revised plans

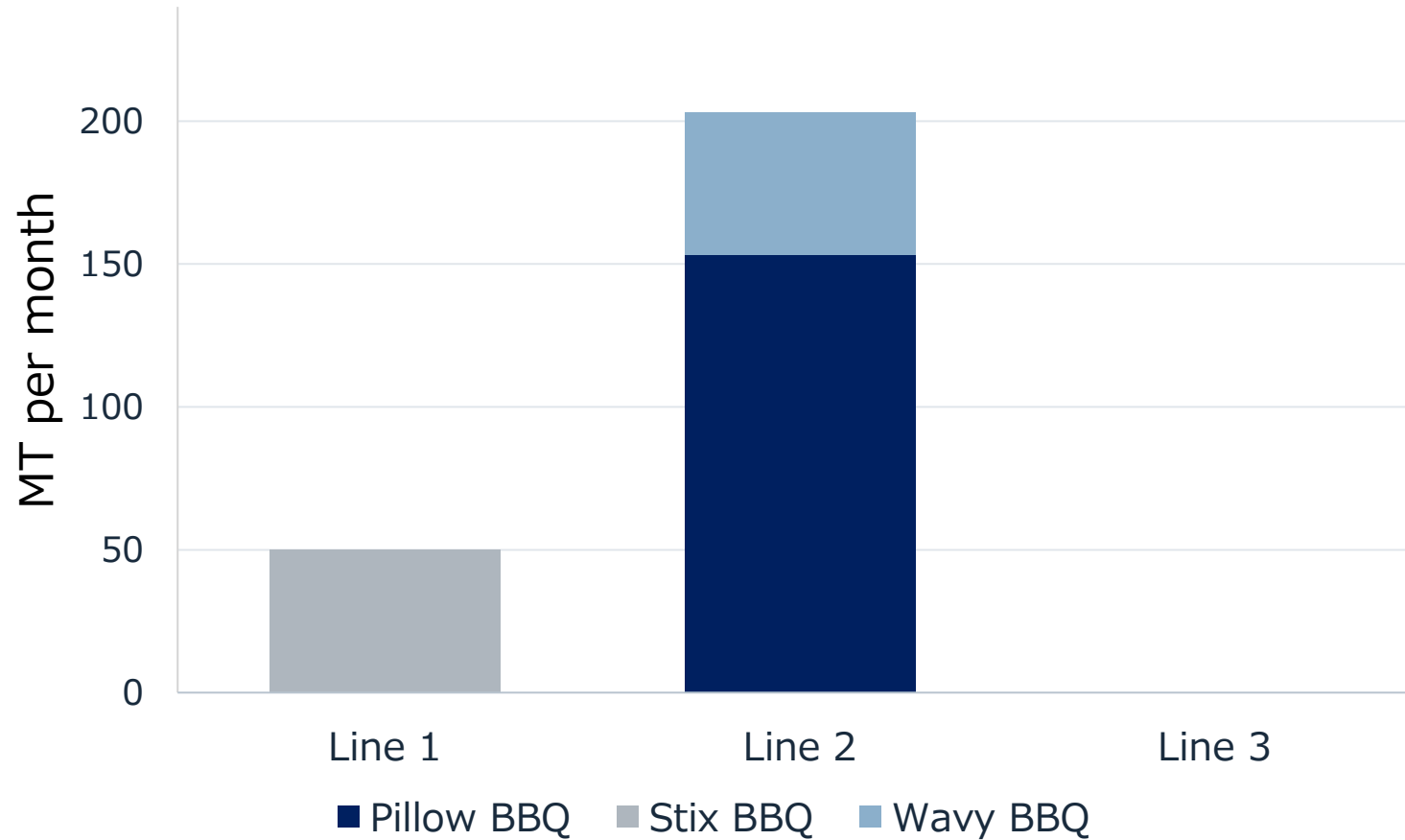
Recorded CBC run: 58 iterations, 0 branch nodes, approximately 0.02 seconds

PuLP formulates the model and the CBC solver returns the optimal production plan.

SKU / operating status	Line 1	Line 2	Line 3	Total MT/day
Pillow BBQ	0.00	6.12	0.00	6.12
Stix BBQ	2.00	0.00	0.00	2.00
Wavy BBQ	0.00	2.00	0.00	2.00
Total MT/day	2.00	8.12	0.00	10.12
Line status	Active	Active	Idle	2 active lines

Line 3 can replace Line 1 without changing modeled cost because their input costs and capacities are identical.

Line 2 produces Pillow and Wavy, while Line 1 produces Stix.



80.24%

Share of volume on Line 2

Line 1: 50 MT Stix

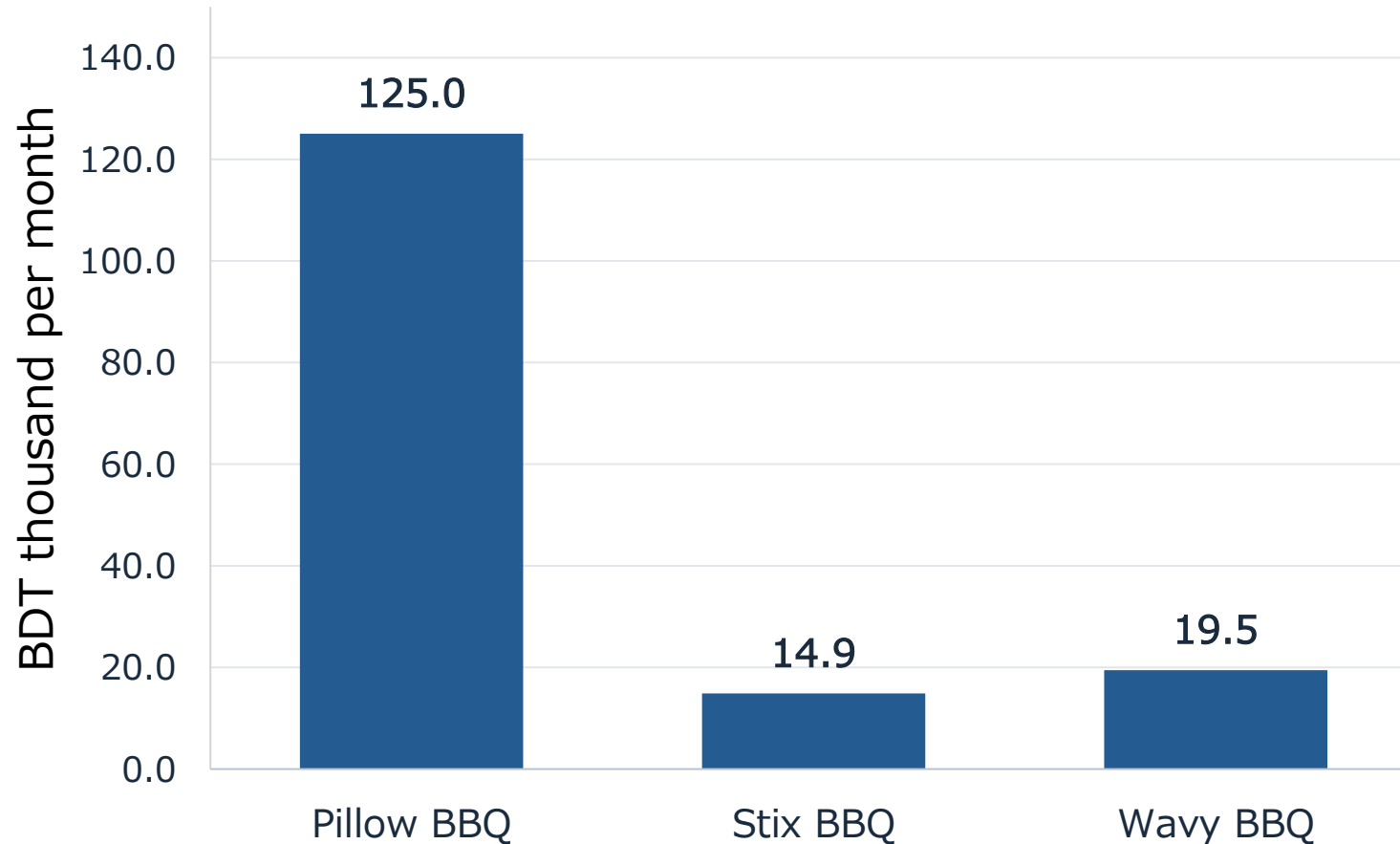
Line 2: 153 MT Pillow and 50 MT Wavy

Line 3: 0 MT

The plan produces 203 MT on Line 2 and 50 MT on Line 1.

Requirement	Plan result	Check
Pillow demand, 153 MT/month	$25 \times 6.12 = 153$ MT	Met exactly
Stix and Wavy, 50 MT/month each	$25 \times 2.00 = 50$ MT each	Met exactly
Line 2 Pillow and Wavy capacity	$6.12 \leq 11.00$ and $2.00 \leq 7.00$ MT/day	Within SKU limits
Line 1 Stix capacity	$2.00 \leq 5.87$ MT/day	Within SKU limit
Maximum 2 active lines	Line 1 and Line 2 active	2 of 2 allowed
Maximum 2 SKUs per line	Line 1: 1 SKU, Line 2: 2 SKUs	Within operating limit

The base solution meets all original demand and operating constraints.



Benefits compare each SKU on Line 2 versus Line 1 or Line 3 at the stated monthly demand.

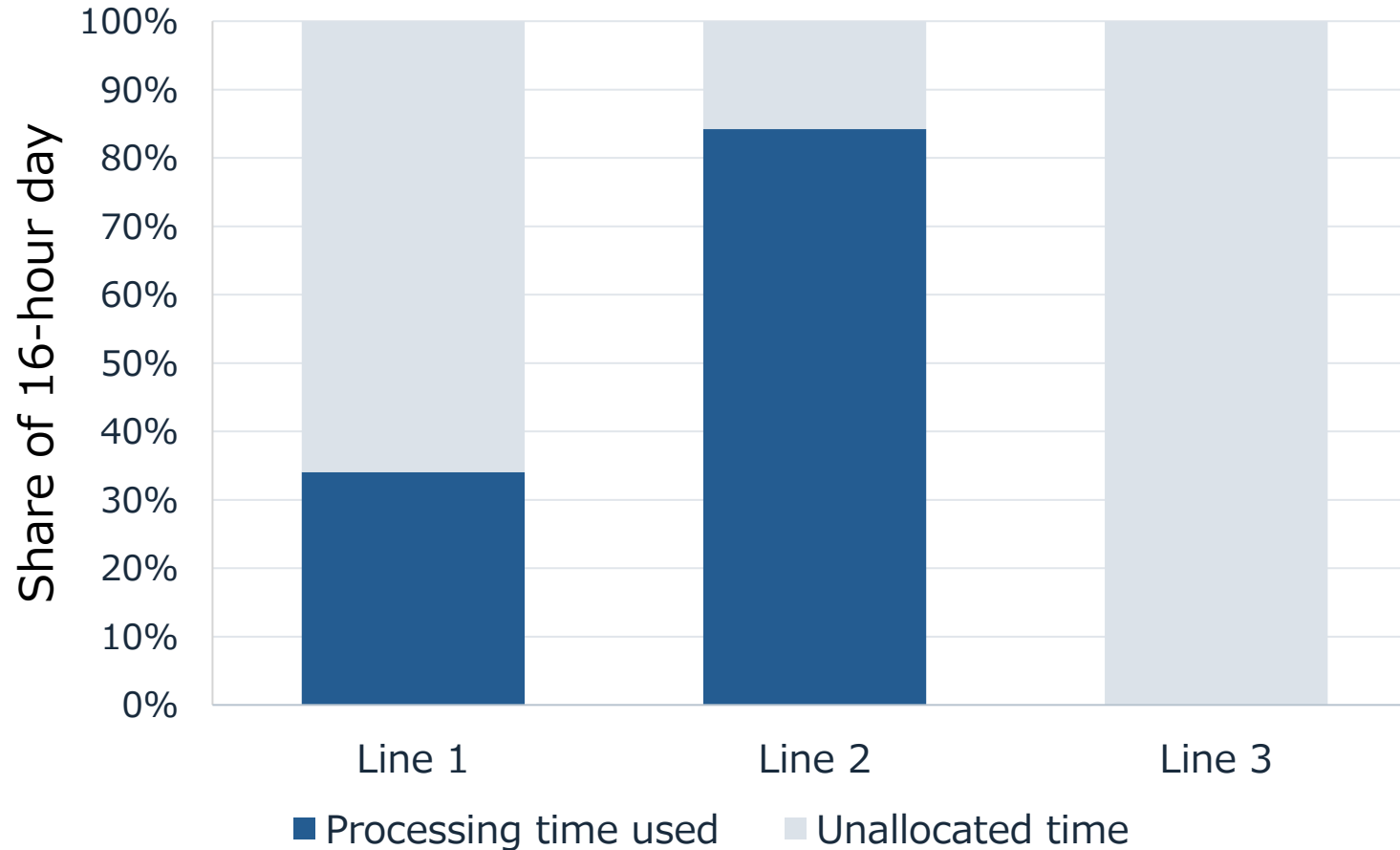
BDT 4,610

More variable-cost benefit from placing Wavy rather than Stix on Line 2

Line 2 can take only 2 SKUs.

Stix uses the smaller line.

Pillow generates the largest variable-cost benefit from using Line 2.



2.53 hours

Line 2 time remaining

Pillow: 8.90 hours

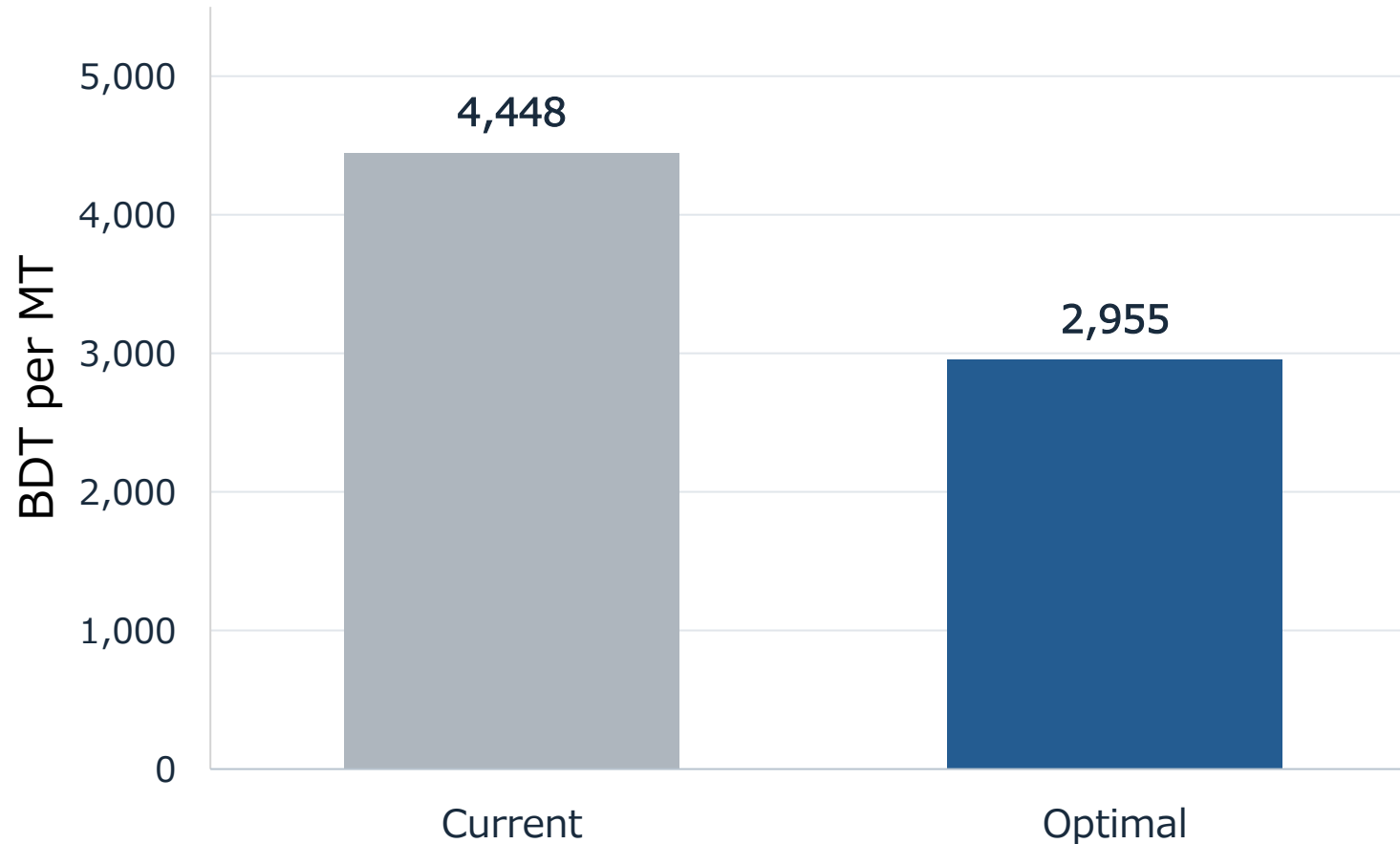
Wavy: 4.57 hours

Line 1 Stix: 5.45 hours

Setup time must fit.

Derived check assumes constant rates and excludes changeovers, cleaning, stoppages and yield loss.

The base allocation uses 84.21% of Line 2's available processing time before changeovers.



BDT 4.53 million

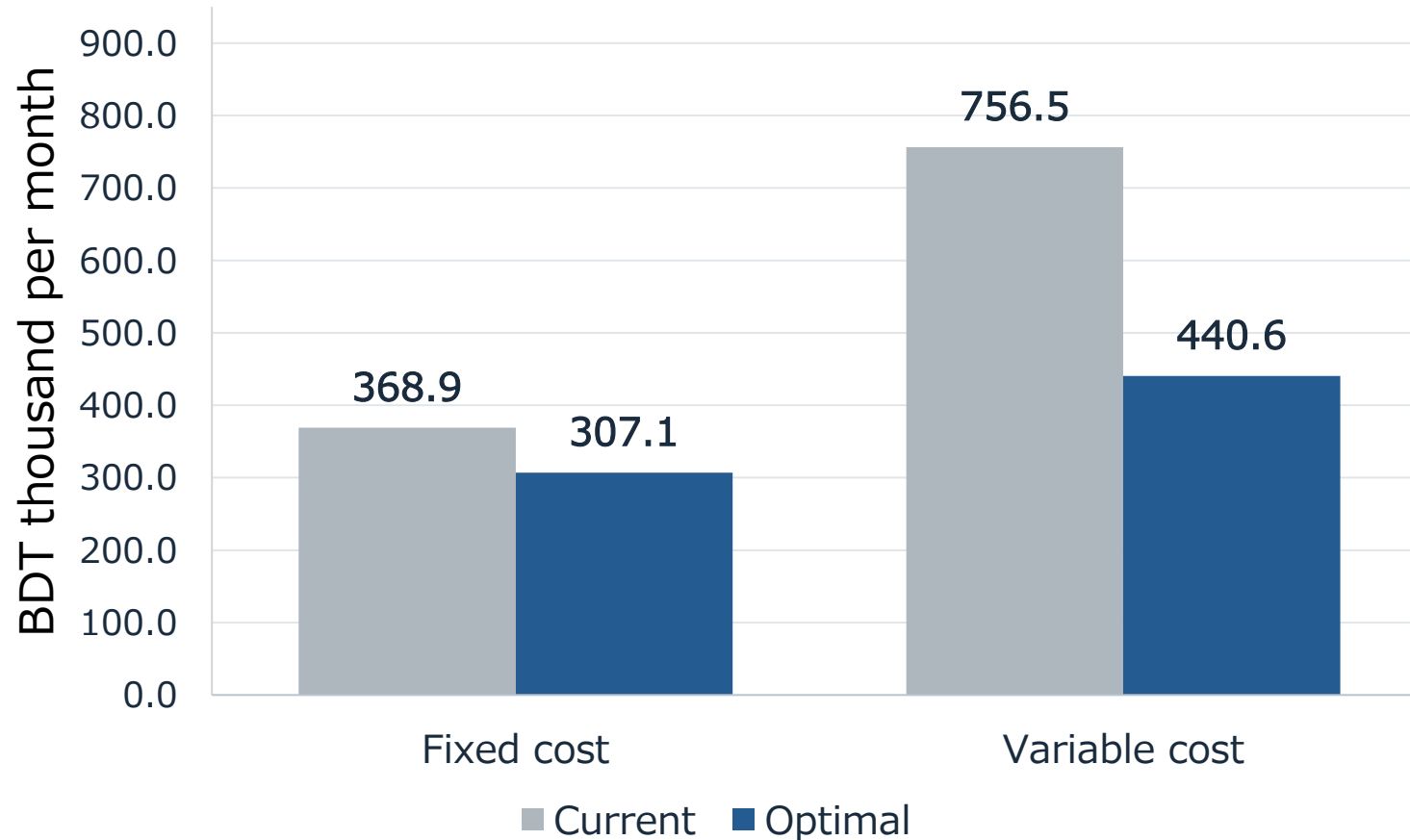
Annualized modeled savings

Monthly savings:
BDT 377,684.58

Same-volume comparison:
253 MT/month

Annualization is illustrative and assumes 12 months at the same demand, cost and realized savings.

Cost per MT falls from BDT 4,448.12 to BDT 2,955.30 at 253 MT/month.



41.76%

Variable cost reduction

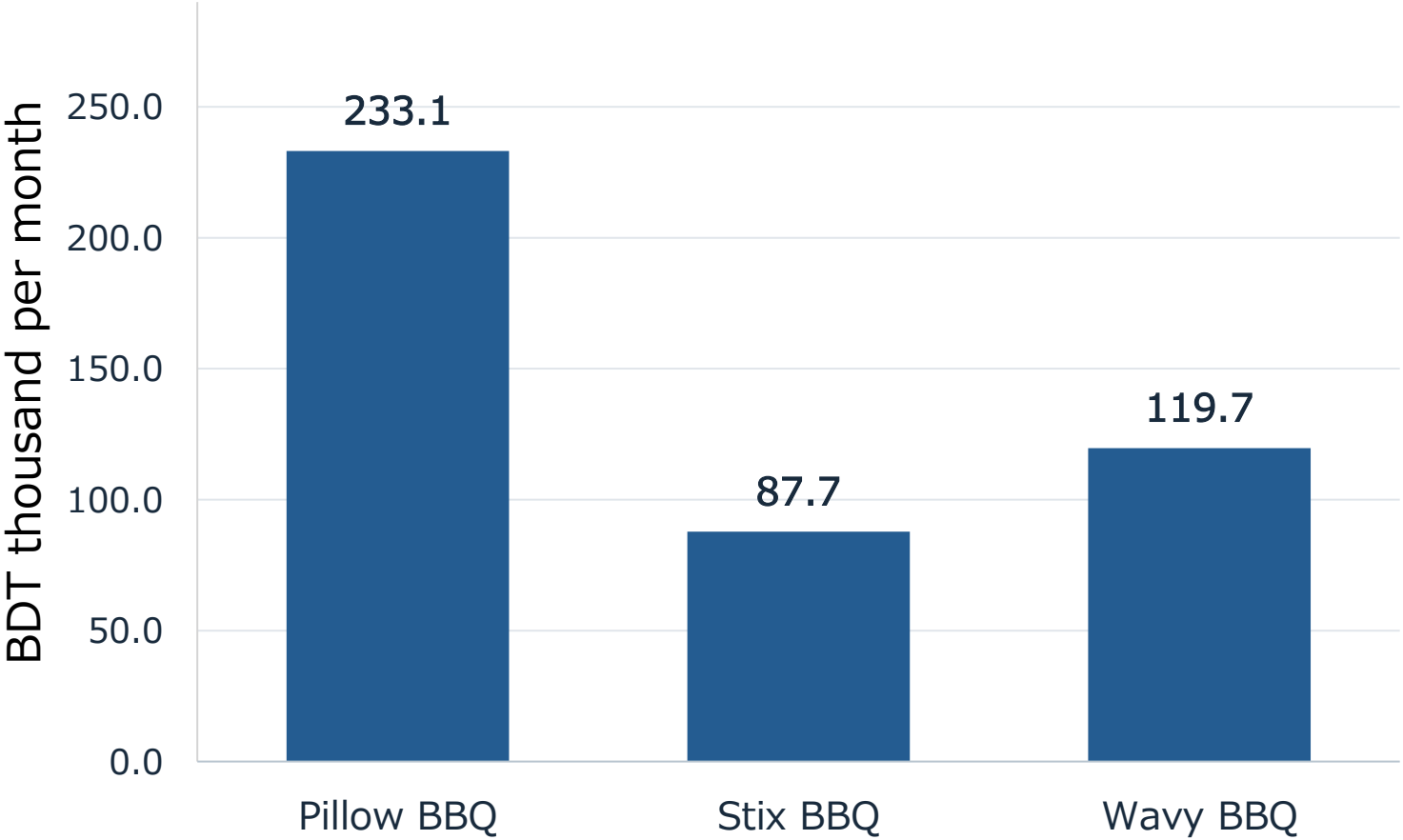
Fixed cost reduction:
16.74%

Fixed savings: BDT
61,750

Variable savings deliver
83.65% of total savings.

Optimal component values reconcile to the objective. Source chart components differ by BDT 250.

Variable cost contributes BDT 315,934.58 of the modeled monthly savings.



BDT 440,565.42

Total variable cost

Pillow: BDT 233,116.92
Stix: BDT 87,734.00
Wavy: BDT 119,714.50

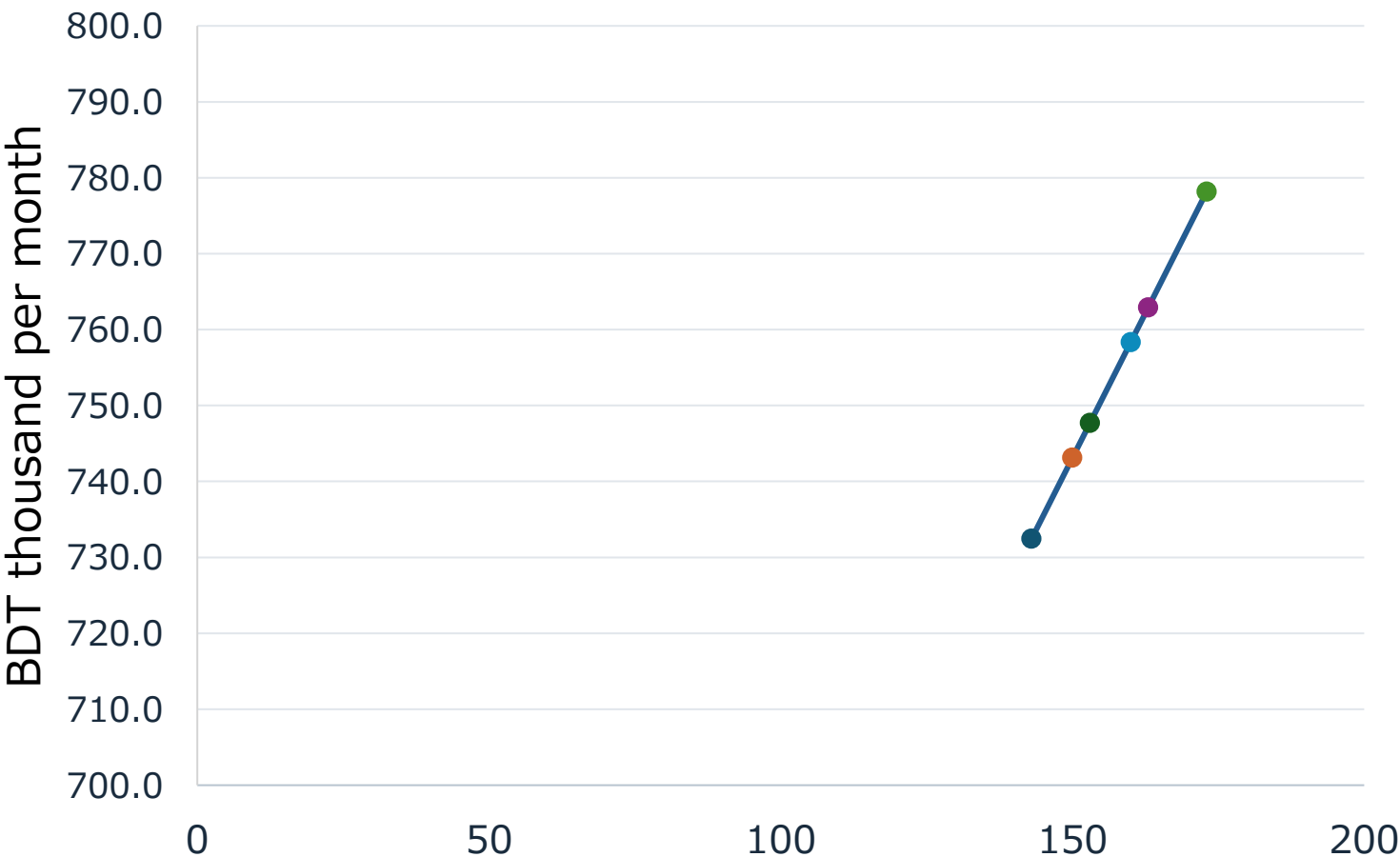
Fixed line costs:
BDT 307,125.00

Pillow represents 52.91% of optimized variable production cost.

Metric	Source chart values	Reconciled calculation
Optimal fixed cost, BDT	307,375.00	307,125.00
Optimal variable cost, BDT	440,315.42	440,565.42
Optimal total cost, BDT	747,690.42	747,690.42
Fixed cost reduction	16.67% reported	16.74%
Variable cost reduction	41.80% reported	41.76%
Total cost reduction	33.56%	33.56%

Reconciled values follow the line fixed costs, production quantities and objective coefficients.

The total optimum remains BDT 747,690.42 after correcting a BDT 250 component mismatch.



+2.04%

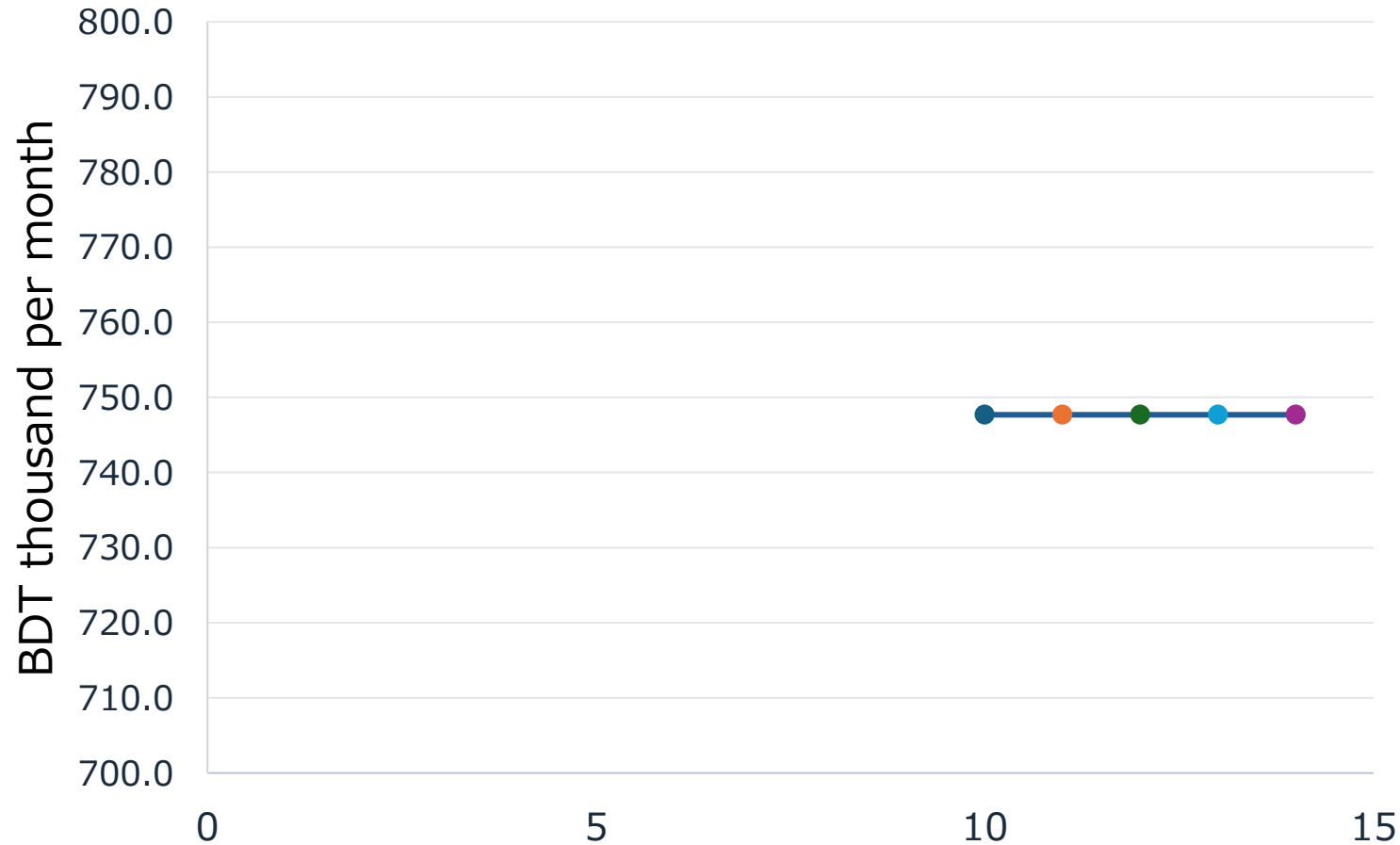
Cost increase for +10 MT from the 153 MT base

153 MT: BDT 747,690.42
163 MT: BDT 762,926.82

Unit increment:
BDT 1,523.64 per MT

Horizontal axis: Pillow monthly demand, MT. Additional points extend the notebook’s 150 and 160 MT tests.

An extra 10 MT of Pillow demand adds BDT 15,236.40 under the unchanged allocation.



No cost change

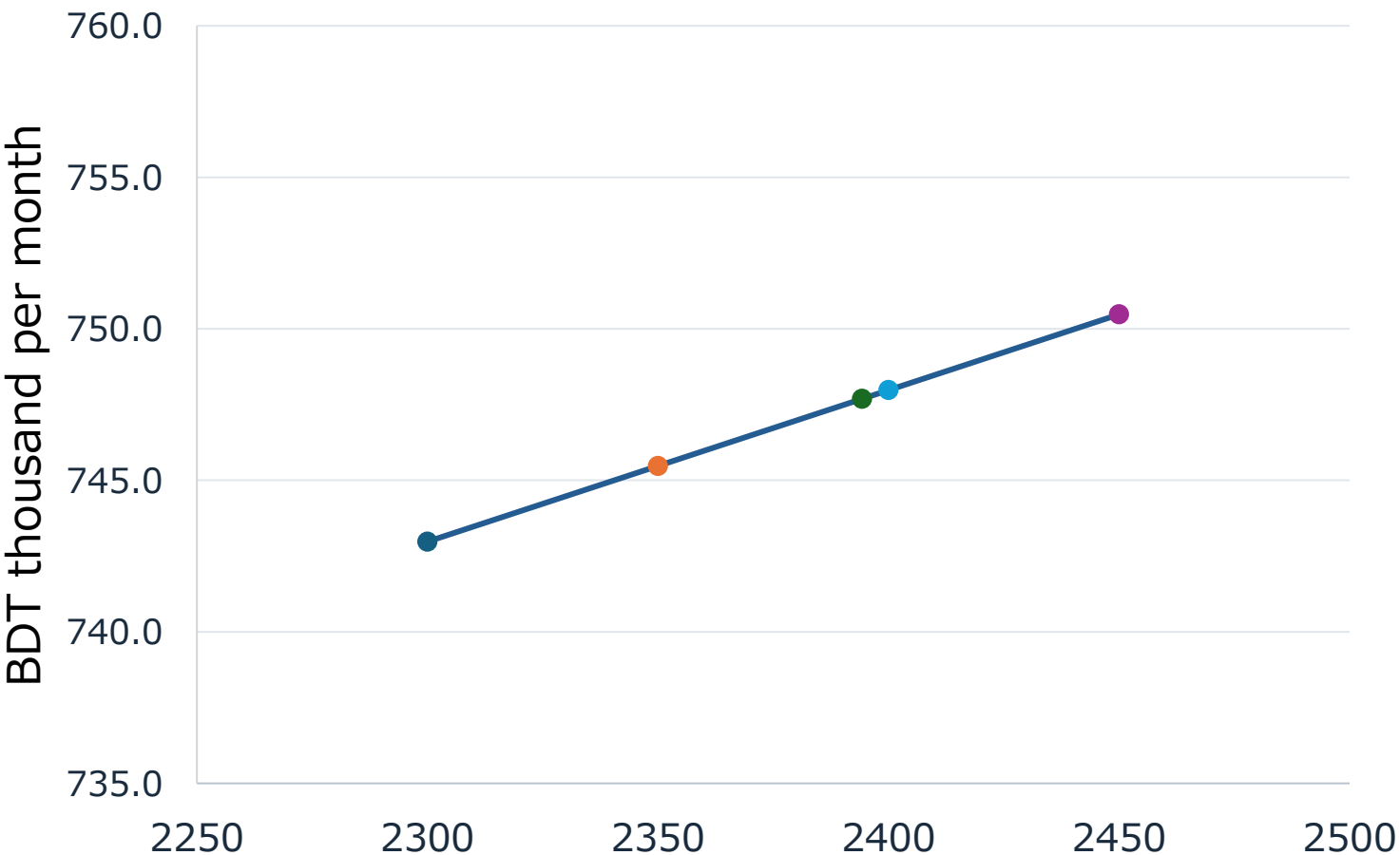
Base Pillow output:
6.12 MT/day

Tested capacity:
10–14 MT/day

Additional Pillow capacity
has no modeled cost
benefit
at current demand.

Horizontal axis: Line 2 Pillow capacity, MT/day. Demand and all other base inputs stay fixed.

Changing Pillow capacity from 10 to 14 MT/day leaves the base optimum unchanged.



+0.33%

Cost increase for BDT 50/MT higher Wavy cost

Wavy volume: 50 MT/month

Allocation stays unchanged across the tested range.

Horizontal axis: Wavy variable cost on Line 2, BDT/MT. Exact base rate is BDT 2,394.29/MT.

An extra BDT 50 per MT of Wavy cost increases monthly cost by BDT 2,500.

Pillow demand	Original model cost	With shared time	Operational implication
153 MT/month	BDT 747,690.42	BDT 747,690.42	Base plan remains feasible
193 MT/month	BDT 808,636.02	BDT 808,636.02	Line 2 nearly fills its available time
203 MT/month	BDT 823,872.42	BDT 825,501.20	About 0.167 MT/day Wavy moves to Line 1

Shared-time constraint: $\sum_s [Q(s,l) / P(s,l)] \leq y(l)$

Additional scenario assumes constant rates and zero changeover time. It is a model extension for validation.

At 203 MT Pillow demand, shared processing time changes the optimal Wavy allocation.

Issue in the supplied model	Recommended refinement	Why it matters
Separate SKU capacity limits	Add shared-time and setup/cleaning constraints	Protect feasibility when SKUs share a line
Fixed cost assumed avoidable	Separate committed and avoidable costs	Validate actual cash savings
Single representative day	Add day-level demand, inventory and maintenance	Allow a workable month schedule
Unused SKU flags can equal 1	Add $z(s,l) \leq y(l)$, and report actual $Q > 0$	Keep assignment reports meaningful
Sensitivity input mutation	Deep-copy or rebuild base data for each scenario	Keep one-factor tests independent

Also correct notebook currency labels from “\$” to BDT and check labor-cost categories for overlap.

Savings require realistic shared capacity and genuinely avoidable line costs.

Phase	Proposed timing	Responsible function	Evidence required
Input confirmation	Week 1	Production and Finance	Demand, rates, cost scope and avoidable costs
Model refinement	Week 2	Planning / analyst	Shared-time, setup, yield and maintenance limits
Controlled line trial	Weeks 3–4	Production and Quality	Actual output, hours, waste and product quality
Pilot review	End of Week 4	Plant head and Finance	Realized savings and operational exceptions
Recurring planning	After acceptance	Planning and Production	Monthly refresh with weekly variance review

Suggested schedule. Pilot release depends on plant readiness and validated operating assumptions.

A controlled pilot can test capacity feasibility and savings before routine use.

KPI	Model reference / acceptance basis	Measurement source
Monthly SKU delivery	153 MT Pillow, 50 MT Stix, 50 MT Wavy	Dispatch and production records
Production-related cost	BDT 747,690.42 modeled reference	Finance ledger and line-level cost log
Daily line output	Line 1: 2.00 MT, Line 2: 8.12 MT	Daily production report
Line 2 occupied time	Processing estimate 13.47 hours plus actual setups	Time study and machine log
Quality and wastage	Meet existing product specifications and plant standards	Quality release and reject log
Realized savings	Comparable baseline minus actual cost minus added charges	Finance-approved pilot reconciliation

The pilot should measure delivery, processing time and realized cost against the model.

Result

Monthly production-related cost falls from BDT 1,125,375.00 to BDT 747,690.42 while meeting 253 MT of demand.

Operating plan

Line 2: Pillow 6.12 MT/day and Wavy 2.00 MT/day

Line 1: Stix 2.00 MT/day. Line 3 remains available as an alternative.

Proposed decision

Proceed to a controlled pilot after validating shared processing time, changeovers, quality and avoidable costs.

The project demonstrates a 33.56% modeled cost reduction with a feasible base allocation.

Variable group	Line 1	Line 2	Line 3	Unit / meaning
QP: Pillow quantity	0.00	6.12	0.00	MT/day
QS: Stix quantity	2.00	0.00	0.00	MT/day
QW: Wavy quantity	0.00	2.00	0.00	MT/day
y: line activation	1	1	0	Binary on/off
Actual produced SKUs	Stix	Pillow and Wavy	None	Read from $Q > 0$

Suggested binary consistency rule: $z(s,l) \leq y(l)$

The production quantities determine actual output, while y identifies active lines.

Calculation	BDT
Pillow variable: $153 \times 1,523.64$	233,116.92
Stix variable: $50 \times 1,754.68$	87,734.00
Wavy variable: $50 \times 2,394.29$	119,714.50
Optimal variable subtotal	440,565.42
Optimal fixed: $102,375 + 204,750$	307,125.00
Optimal total: variable + fixed	747,690.42
Savings: $1,125,375.00 - 747,690.42$	377,684.58

$$\text{Reduction} = 377,684.58 \div 1,125,375.00 \times 100 = 33.56\%$$

The reconciled fixed and variable costs add exactly to the optimized objective.

Source / evidence type	Use in this report
Research paper by Md Tanvir Hasan	Plant context, model inputs, baseline and study conclusions
Python notebook, cells 0 and 2	Model formulation, recorded solver output and sensitivity cases
Notebook cost charts, cells 20–26	Baseline comparison with a reconciled optimal breakdown
Additional calculations	Daily/monthly conversion, cost per MT and shared-time scenarios
Proposed actions	Pilot schedule, model refinements and operating KPIs
Sample PPT and material guidance	Original bands, dimensions, typography and restrained color use

Savings are model estimates. Additional scenarios and proposed actions require operational validation.

The report distinguishes supplied project results from additional calculations and proposals.

Thank You

Questions and discussion

Md Tanvir Hasan